

問題 329

$0 \leq \theta < 2\pi$ で $\sin \theta + \sqrt{3} \cos \theta = -1$ を解け.

解答:

$$\begin{aligned}\sin \theta + \sqrt{3} \cos \theta &= 2 \sin \left(\theta + \frac{\pi}{3} \right) = -1 \\ \sin \left(\theta + \frac{\pi}{3} \right) &= -\frac{1}{2} \\ \theta + \frac{\pi}{3} &= \frac{7\pi}{6}, \frac{11\pi}{6} \\ \theta &= \frac{5\pi}{6}, \frac{3\pi}{2}\end{aligned}$$

問題 330

$0 \leq \theta < 2\pi$ のとき, $y = \sin \theta - \sqrt{3} \cos \theta$ の最大値と最小値を求めよ.

解答:

$$\begin{aligned}y &= 2 \sin \left(\theta - \frac{\pi}{3} \right) \\ \text{最大値 } 2 \left(\theta - \frac{\pi}{3} = \frac{\pi}{2}, \theta = \frac{5\pi}{6} \right), &\text{ 最小値 } -2 \left(\theta = \frac{11\pi}{6} \right)\end{aligned}$$

問題 331

$\tan \alpha + \tan \beta = \frac{\sin(\alpha + \beta)}{\cos \alpha \cos \beta}$ を証明せよ.

解答:

$$\begin{aligned}\text{左辺} &= \frac{\sin \alpha}{\cos \alpha} + \frac{\sin \beta}{\cos \beta} = \frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\cos \alpha \cos \beta} = \\ \frac{\sin(\alpha + \beta)}{\cos \alpha \cos \beta} &= \text{右辺} \quad \square\end{aligned}$$

問題 332

$\sin(\alpha + \beta) \sin(\alpha - \beta) = \sin^2 \alpha - \sin^2 \beta$ を証明せよ.

解答:

$$\begin{aligned}\text{左辺} &= (\sin \alpha \cos \beta + \cos \alpha \sin \beta)(\sin \alpha \cos \beta - \cos \alpha \sin \beta) \\ &= \sin^2 \alpha \cos^2 \beta - \cos^2 \alpha \sin^2 \beta \\ &= \sin^2 \alpha (1 - \sin^2 \beta) - (1 - \sin^2 \alpha) \sin^2 \beta \\ &= \sin^2 \alpha - \sin^2 \alpha \sin^2 \beta - \sin^2 \beta + \sin^2 \alpha \sin^2 \beta = \\ \sin^2 \alpha - \sin^2 \beta &= \text{右辺} \quad \square\end{aligned}$$

問題 333

2 直線 $y = \frac{1}{2}x + 1$, $y = 2x - 3$ のなす角 θ を求めよ.

解答:

$$\begin{aligned}\text{傾き } m_1 &= \frac{1}{2}, \quad m_2 = 2 \text{ とすると} \\ \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{\frac{1}{2} - 2}{1 + 1} \right| = \left| \frac{-\frac{3}{2}}{2} \right| = \frac{3}{4}\end{aligned}$$

$$\theta = \arctan \frac{3}{4} \approx 36.87^\circ$$

問題 334

$\sin 2\theta = 2 \sin \theta \cos \theta$ と $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ を用いて $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$ を導け.

解答:

$$\begin{aligned}\tan 2\theta &= \frac{\sin 2\theta}{\cos 2\theta} = \frac{2 \sin \theta \cos \theta}{\cos^2 \theta - \sin^2 \theta} \\ \text{分子分母を } \cos^2 \theta &\text{ で割る:} \\ &= \frac{2 \frac{\sin \theta}{\cos \theta}}{1 - \frac{\sin^2 \theta}{\cos^2 \theta}} = \frac{2 \tan \theta}{1 - \tan^2 \theta} \quad \square\end{aligned}$$

問題 335

$\cos^4 \theta - \sin^4 \theta = \cos 2\theta$ を示せ.

解答:

$$\begin{aligned}\text{左辺} &= (\cos^2 \theta + \sin^2 \theta)(\cos^2 \theta - \sin^2 \theta) = 1 \cdot \cos 2\theta = \\ \text{右辺} & \quad \square\end{aligned}$$

問題 336

$0 \leq \theta < 2\pi$ のとき $2 \cos^2 \theta + \sin \theta - 1 = 0$ を解け.

解答:

$$\begin{aligned}\cos^2 \theta &= 1 - \sin^2 \theta \text{ を代入:} \\ 2(1 - \sin^2 \theta) + \sin \theta - 1 &= 0, \quad -2 \sin^2 \theta + \sin \theta + 1 = 0 \\ 2 \sin^2 \theta - \sin \theta - 1 &= 0, \quad (2 \sin \theta + 1)(\sin \theta - 1) = 0 \\ \sin \theta &= -\frac{1}{2} \text{ または } \sin \theta = 1 \\ \sin \theta = 1 : \theta &= \frac{\pi}{2} \\ \sin \theta = -\frac{1}{2} : \theta &= \frac{7\pi}{6}, \frac{11\pi}{6} \\ \therefore \theta &= \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}\end{aligned}$$